

```
In [1]: pip install ultralytics opencv-python matplotlib numpy

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```
In [3]: import cv2
import numpy as np
import matplotlib.pyplot as plt

from ultralytics import YOLO
```

Creating new Ultralytics Settings v0.0.8 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com/usage/settings>.

```
In [4]: model = YOLO("yolov8n.pt")
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolov8n.pt': 100% ————— 6.2MB 71.0MB/s 0.1s

```
In [5]: image_path = "image.jpg"

image = cv2.imread(image_path)

if image is None:
    print("Image not found!")
else:
    print("Image loaded successfully!")
```

Image loaded successfully!

```
In [6]: image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(10, 6))
plt.imshow(image_rgb)
plt.axis("off")
```

```
plt.title("Original Image")
plt.show()
```

Original Image



```
In [7]: results = model(image)
```

0: 640x640 1 person, 530.0ms

Speed: 21.5ms preprocess, 530.0ms inference, 42.1ms postprocess per image at shape (1, 3, 640, 640)

```
In [8]: result_image = results[0].plot()
```

```
result_image = cv2.cvtColor(result_image, cv2.COLOR_BGR2RGB)
```

```
plt.figure(figsize=(12, 8))
plt.imshow(result_image)
plt.axis("off")
plt.title("YOLOv8 Object Detection")
plt.show()
```

YOLOv8 Object Detection



```
In [9]: result = results[0]
```

```
for box in result.bboxes:
```

```

class_id = int(box.cls[0])
confidence = float(box.conf[0])

class_name = model.names[class_id]

print(
    f"Object: {class_name}, "
    f"Confidence: {confidence:.2f}"
)

```

Object: person, Confidence: 0.50

```

In [10]: for box in result.boxes:
          x1, y1, x2, y2 = box.xyxy[0].tolist()

          class_id = int(box.cls[0])
          confidence = float(box.conf[0])
          class_name = model.names[class_id]

          print(f"Object: {class_name}")
          print(f"Confidence: {confidence:.2f}")
          print(f"Bounding Box: ({x1:.1f}, {y1:.1f}, {x2:.1f}, {y2:.1f})")
          print()

```

Object: person
Confidence: 0.50
Bounding Box: (102.0, 74.1, 539.4, 623.9)

```

In [11]: results[0].save(filename="detected_output.jpg")
          print("Detection result saved successfully.")

```

Detection result saved successfully.

15. How YOLOv8 Works

YOLO stands for:

You Only Look Once

YOLO performs object detection in a single neural network inference.

Basic workflow:

Input Image ↓ Image Preprocessing ↓ YOLOv8 Neural Network ↓ Feature Extraction ↓ Object Detection ↓ Bounding Boxes ↓ Class Labels + Confidence Scores ↓ Non-Maximum Suppression ↓ Final Detection

YOLO can detect multiple objects simultaneously.

16. Important Terms Bounding Box

A bounding box is a rectangular box around a detected object.

It is usually represented as:

(x1, y1, x2, y2)

It tells us where the object is located in the image.

Confidence Score

The confidence score indicates how confident the model is about a detection.

For example:

person → 0.92

means the model has a confidence score of approximately 92% for that detection.

IoU

IoU stands for:

Intersection over Union

It measures the overlap between the predicted bounding box and the ground-truth bounding box.

Formula:

$$IoU = \frac{\text{Area of Intersection}}{\text{Area of Union}}$$

A higher IoU indicates better overlap.

- ## 17. Questions and Answers Q1. Differentiate between image classification, object detection, and image segmentation. Answer: Task Description
- Image Classification Assigns one or more labels to the entire image
Object Detection Detects objects and provides bounding boxes and class labels
Image Segmentation Assigns a class to individual pixels
Example:

For an image containing a dog and a person:

Classification:

Dog + Person

Object Detection:

Dog → Bounding Box Person → Bounding Box

Segmentation:

Dog → Pixel-level mask Person → Pixel-level mask

Q2. Explain the working principle of the YOLO object detection algorithm. Answer:

YOLO treats object detection as a single regression problem.

The input image is given to a deep neural network. The network extracts features and predicts:

Bounding box coordinates Object classes Confidence scores

The predictions are then processed using Non-Maximum Suppression (NMS) to remove duplicate overlapping detections.

Thus, YOLO can perform object detection in a single forward pass, making it suitable for real-time applications.

Q3. Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and practical applications. Answer:

Feature	YOLO	SSD	Faster R-CNN
Speed	Very fast	Fast	Relatively slower
Accuracy	High	High	Generally high
Architecture	One-stage	One-stage	Two-stage
Real-time use	Excellent	Good	More limited
Complexity	Moderate	Moderate	Higher
Applications	Real-time detection, surveillance	Mobile/real-time applications	High-accuracy detection

YOLO and SSD are one-stage detectors, while Faster R-CNN is a two-stage detector.

Q4. What is a bounding box? Why is it important? Answer:

A bounding box is a rectangular region surrounding a detected object.

It is important because it provides the location and size of the object within an image.

Example:

Object: Car

(x1,y1) _____ | CAR | | | _____ (x2,y2)

Confidence Score

Q5. Explain the significance of confidence score and IoU. Answer:

It represents how confident the model is that a detected region belongs to a particular object class.

IoU

IoU measures the overlap between the predicted bounding box and the actual ground-truth bounding box.

$$IoU = \frac{Intersection}{Union}$$

Both are important for evaluating object detection performance.

Q6. What are the advantages of using pre-trained deep learning models over traditional image processing techniques? Answer:

Pre-trained deep learning models have several advantages:

They can recognize complex objects. They automatically learn useful features. They require less manual feature engineering. They can be fine-tuned for specific applications. They generally perform better on complex images. They can detect multiple object categories. They can be used for real-time applications.

Q7. Why are datasets such as MS COCO and Pascal VOC widely used? Answer:

MS COCO and Pascal VOC are widely used because they provide:

Large collections of images. Multiple object categories. Bounding-box annotations. Standardized evaluation methods. Benchmark datasets for comparing detection algorithms.

They help researchers evaluate and compare different object detection models.

Q8. Mention five real-world applications of object detection. Answer:

Five applications are:

Autonomous vehicles – detecting cars, pedestrians, traffic signs, etc. Surveillance systems – detecting people and vehicles. Healthcare – detecting abnormalities in medical images. Retail analytics – detecting products and monitoring shelves. Industrial inspection – detecting defects in manufactured products.

Other applications include:

Agriculture Robotics Smart cities Traffic monitoring Security systems

Q9. What challenges are commonly encountered while detecting objects in complex real-world environments? Answer:

Common challenges include:

Poor lighting conditions. Object occlusion. Small objects. Objects with similar appearance. Complex backgrounds. Different object orientations. Different scales of objects. Motion blur. Low-resolution images. Real-time processing requirements.

Q10. How can object detection systems be further improved using recent advancements? Answer:

Object detection systems can be improved using:

Better deep learning architectures. Transformer-based vision models. Improved data augmentation. Larger and more diverse datasets. Transfer learning. Model pruning and quantization. GPU acceleration. Better feature extraction techniques. Multi-scale detection. Improved loss functions and training strategies.

These techniques can improve accuracy, robustness, and inference speed.

18. Advantages of YOLOv8 Fast inference. Suitable for real-time applications. Can detect multiple objects. Provides bounding boxes and confidence scores. Easy to implement using Ultralytics. Supports GPU acceleration. Can be trained on custom datasets. Useful for surveillance, robotics, autonomous systems, etc.

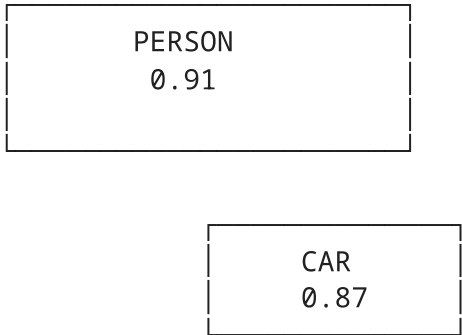
19. Limitations

Despite its advantages, object detection has some limitations:

Performance can decrease for very small objects. Detection can be affected by poor lighting. Occluded objects may be missed. High-resolution images require more computational resources. Pre-trained models may not recognize domain-specific objects. Accuracy depends on training data.

20. Expected Output

After running the program, the output image should look approximately like:



The actual output depends on the image you provide.

21. Observation

The YOLOv8 model successfully detects multiple objects present in the input image. For each detected object, the model provides:

Object class Bounding box Confidence score

The model performs detection in a single inference process, making it suitable for real-time object detection applications.

22. Result

The object detection and recognition system was successfully implemented using the pre-trained YOLOv8 deep learning model. Multiple objects were detected and localized using bounding boxes, class labels, and confidence scores.

23. Conclusion

In this experiment, we studied and implemented deep learning-based object detection using YOLOv8. We learned how object detection differs from image classification and segmentation. The experiment demonstrated how a pre-trained model can identify and localize multiple objects in an image.

YOLOv8 provides a good combination of detection accuracy and processing speed, making it useful for applications such as autonomous vehicles, surveillance, robotics, healthcare, and smart-city systems.